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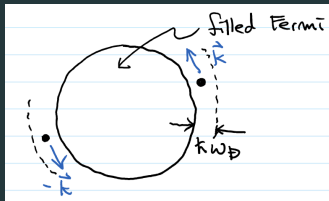
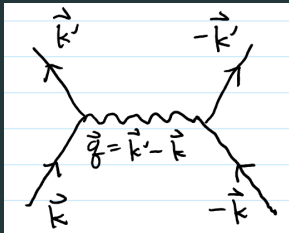
Superconductivity

Lecture 19 – “Ground for Many”

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Previously, electron pairing via lattice



$$E \simeq 2E_F - 2\hbar\omega_D e^{-2/D(E_F)V} \quad (1)$$

- Bound state exists;
- SC energy scale set by θ_D ;
- $T_c \sim 1$ K due to exponential term.

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From two to many

Second quantisation representation

Quantum mechanical operators:

- create a particle by $c^\dagger$
- destroy a particle by c .

To create or destroy an electron with momentum $\mathbf{k}$ and spin σ

- $c_{\mathbf{k}\sigma}^\dagger$
- $c_{\mathbf{k}\sigma}$

Second quantisation rules

Fermions respecting anti-commutativity:

$$\{c_{k\sigma}, c_{k'\sigma'}^\dagger\} \equiv c_{k\sigma} c_{k'\sigma'}^\dagger + c_{k'\sigma'}^\dagger c_{k\sigma} \quad (2)$$

$$= \delta_{kk'} \delta_{\sigma\sigma'}, \quad (3)$$

and

$$\{c_{k\sigma}, c_{k'\sigma'}\} = \{c_{k\sigma}^\dagger, c_{k'\sigma'}^\dagger\} \quad (4)$$

$$= 0. \quad (5)$$

The number operator

$$n_{k\sigma} \equiv c_{k\sigma}^\dagger c_{k\sigma}$$

returns the number of particles having momentum k and spin σ

From vacuum to BCS

Denote vacuum by $|0\rangle$.

Fermi sea:

$$|FS\rangle = \prod_{k \leq k_F} c_{k\uparrow}^\dagger c_{k\downarrow}^\dagger |0\rangle. \quad (6)$$

obtained by adding to the vacuum one electron after another, from $k = 0$ all the way up to $k = k_F$ (for both spin up and down).

To create a pair:

$$c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger |0\rangle. \quad (7)$$

To create **many** pairs, Bardeen, Schrieffer and Cooper attempted:

$$|BCS\rangle = \prod_k \left(u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \right) |0\rangle, \quad (8)$$

BCS ground state

$$|BCS\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} \right) |0\rangle, \quad (9)$$

where $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$, and the sum over all $\mathbf{k}$ values (M) in the band.

- $|v_{\mathbf{k}}|^2$ the probability of $(\mathbf{k} \uparrow, -\mathbf{k} \downarrow)$ occupied;
- $|u_{\mathbf{k}}|^2$ the probability of $(\mathbf{k} \uparrow, -\mathbf{k} \downarrow)$ unoccupied.

Any N possible from 0 to $2M \leftrightarrow$ particle number can vary.

However, N is sharply centred around $\bar{N}$, i.e.,

$$\frac{\delta N_{\text{rms}}}{\bar{N}} \sim 10^{-13}, \quad (10)$$

one can show if $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ determined (how?).

One can show (p.51 of Tinkham) the average particle number

$$\bar{N} = \sum_k 2|v_k|^2 \quad (11)$$

and the variation

$$\langle (N - \bar{N})^2 \rangle = \sum_k 4|u_k|^2 |v_k|^2. \quad (12)$$

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Number and phase

$$|BCS\rangle = \prod_{\mathbf{k}} \left(|u_{\mathbf{k}}| + |v_{\mathbf{k}}| e^{i\phi} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} \right) |0\rangle \quad (13)$$

$$= \sum_N \lambda_N |\psi_N\rangle \quad (14)$$

$|\psi_N\rangle$ would have a phase factor $e^{iN\phi/2}$.

One may project out the N -particle part,

$$|\psi_N\rangle = \int_0^{2\pi} d\phi e^{-iN\phi/2} |BCS\rangle \quad (15)$$

Then, one may say

$$\Delta N \Delta \phi \gtrsim 1 \quad (16)$$

Variational method

How to find $|BCS\rangle$

So, how to determine u_k and v_k . How?

Find the minimum

$$\delta E = \delta \langle BCS | \mathcal{H} | BCS \rangle = 0. \quad (17)$$

Or better

$$\delta \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle = 0, \quad (18)$$

where $-\mu N_{op}$ to regulate the varying number of particles and make the the reference of kinetic energy as μ .

Need to write down $\mathcal{H}$...

Pairing Hamiltonian (1/2)

$$\mathcal{H} = \underbrace{\sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} n_{\mathbf{k}\sigma}}_{K.E.} - |V| \underbrace{\sum_{\mathbf{k}, \mathbf{k}'} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}}_{interaction}, \quad (19)$$

Note

- essential terms for superconductivity only;
- terms leading to zero expectation dropped off.

This

$$\delta \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle = 0, \quad (20)$$

becomes

$$\delta \langle BCS | \sum_{\mathbf{k}, \sigma} (\epsilon_{\mathbf{k}} - \mu) n_{\mathbf{k}\sigma} - |V| \sum_{\mathbf{k}, \mathbf{k}'} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} | BCS \rangle = 0, \quad (21)$$

Pairing Hamiltonian (2/2)

$$\langle BCS | \sum_{\mathbf{k}, \sigma} (\epsilon_{\mathbf{k}} - \mu) n_{\mathbf{k}\sigma} - |V| \sum_{\mathbf{k}\mathbf{k}'} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} | BCS \rangle \quad (22)$$

This represents $\langle K.E. - \mu N \rangle + \langle P.E. \rangle$, where

$$\langle K.E. - \mu N \rangle = 2 \sum_{\mathbf{k}} (\epsilon_{\mathbf{k}} - \mu) |v_{\mathbf{k}}|^2 \quad (23)$$

$$\langle P.E. \rangle = -|V| \sum_{\mathbf{k}\mathbf{k}'} u_{\mathbf{k}} v_{\mathbf{k}}^* u_{\mathbf{k}'}^* v_{\mathbf{k}'} \quad (24)$$

which one can show by direct evaluation.

Variational method (1/2)

Now show

$$\delta \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle = 0, \quad (25)$$

under the constraint of $|u_k|^2 + |v_k|^2 = 1$.

$$\langle \dots \rangle = 2 \sum_k (\epsilon_k - \mu) |v_k|^2 - |V| \sum_{kk'} u_k v_k^* u_{k'}^* v_{k'}. \quad (26)$$

Let

$$u_k = \sin \theta_k \quad \text{and} \quad v_k = \cos \theta_k.$$

Then one can show

$$\begin{aligned} \frac{\partial}{\partial \theta_k} \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle &= 0 \\ &= -2(\epsilon_k - \mu) \sin 2\theta_k - |V| \sum_{k'} \cos 2\theta_k \sin 2\theta_{k'} \end{aligned}$$

Variational method (2/2)

Rearranging gives

$$\tan 2\theta_k = -|V| \frac{\sum_{k'} \sin 2\theta_{k'}}{2(\epsilon_k - \mu)}, \quad (27)$$

which, once defining

$$\Delta = \frac{1}{2} |V| \sum_{k'} \sin 2\theta_{k'}, \quad (28)$$

can be rewritten as

$$\tan 2\theta_k = -\frac{\Delta}{(\epsilon_k - \mu)} \quad (29)$$

Now one can show

$$\begin{aligned} \cdot \quad 2u_k v_k &= \sin 2\theta_k = \frac{\Delta}{\sqrt{(\epsilon_k - \mu)^2 + |\Delta|^2}} \equiv \frac{\Delta}{E_k} \\ \cdot \quad v_k^2 - u_k^2 &= \cos 2\theta_k = -\frac{\epsilon_k - \mu}{\sqrt{(\epsilon_k - \mu)^2 + |\Delta|^2}} \equiv -\frac{\epsilon_k - \mu}{E_k} \end{aligned}$$

Gap equation

It follows

$$\Delta = \frac{1}{2}|V| \sum_{k'} \sin 2\theta_{k'} \quad (30)$$

$$= -\frac{1}{2}|V| \sum_{k'} \frac{\Delta}{E_{k'}} \quad (31)$$

That is,

$$1 = \frac{|V|}{2} \sum_k \frac{1}{E_k} \quad (32)$$

where

$$E_k = \sqrt{(\epsilon_k - \mu)^2 + |\Delta|^2}. \quad (33)$$

Replacing the summation by an integration from $-\hbar\omega_D$ to $\hbar\omega_D$

$$\frac{1}{NV} = \int_0^{\hbar\omega_D} \frac{d(\epsilon_k - \mu)}{(\Delta^2 + (\epsilon_k - \mu)^2)^{1/2}} = \sinh^{-1} \frac{\hbar\omega_D}{\Delta} \quad (34)$$

Gap and coefficients

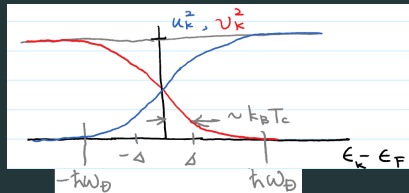
$$\Delta \simeq 2\hbar\omega_D e^{-1/NV} \quad (35)$$

The coefficients can now be written as

$$|u_k|^2 = \frac{1}{2} \left(1 + \frac{\epsilon_k - \mu}{E_k} \right), \quad (36)$$

$$|v_k|^2 = \frac{1}{2} \left(1 - \frac{\epsilon_k - \mu}{E_k} \right). \quad (37)$$

At zero temperature!



Ground state energy

Ground state energy

We have worked on finding the condition (u_k and v_k) for energy minimisation,

$$\delta \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle = 0, \quad (38)$$

What about the ground state energy itself?

$$\langle E \rangle_s = \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle = \sum_k \left((\epsilon_k - \mu) - \frac{(\epsilon_k - \mu)^2}{E_k} \right) - \frac{\Delta^2}{|V|}, \quad (39)$$

while for normal state

$$\langle E \rangle_n = \sum_{|k| < k_F} 2(\epsilon_k - \mu) \quad (40)$$

Thus

$$\langle E \rangle_s - \langle E \rangle_n = 2 \sum_{k > k_F} \left((\epsilon_k - \mu) - \frac{(\epsilon_k - \mu)^2}{E_k} \right) - \frac{\Delta^2}{V} \quad (41)$$